Clinical Infectious Diseases

MAJOR ARTICLE

Sera Neutralizing Activities Against Severe Acute Respiratory Syndrome Coronavirus 2 and Multiple Variants 6 Months After Hospitalization for Coronavirus Disease 2019

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(See the Editorial Commentary by Pirofski on pages e1345–7.)

Background. Humoral response to severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) occurs within the first weeks after coronavirus disease 2019 (COVID-19). Those antibodies exert a neutralizing activity against SARS-CoV-2, whose evolution over time after COVID-19 as well as efficiency against novel variants are poorly characterized.

Methods. In this prospective study, sera of 107 patients hospitalized with COVID-19 were collected at 3 and 6 months postinfection. We performed quantitative neutralization experiments on top of high-throughput serological assays evaluating anti-spike (S) and anti-nucleocapsid (NP) immunoglobulin G (IgG).

Results. Levels of seroneutralization and IgG rates against the ancestral strain decreased significantly over time. After 6 months, 2.8% of the patients had a negative serological status for both anti-S and anti-NP IgG. However, all sera had a persistent and effective neutralizing effect against SARS-CoV-2. IgG levels correlated with seroneutralization, and this correlation was stronger for anti-S than for anti-NP antibodies. The level of seroneutralization quantified at 6 months correlated with markers of initial severity, notably admission to intensive care units and the need for mechanical invasive ventilation. In addition, sera collected at 6 months were tested against multiple SARS-CoV-2 variants and showed efficient neutralizing effects against the D614G, B.1.1.7, and P.1 variants but significantly weaker activity against the B.1.351 variant.

Conclusions. Decrease in IgG rates and serological assays becoming negative did not imply loss of neutralizing capacity. Our results indicate a sustained humoral response against the ancestral strain and the D614G, B.1.1.7, and P.1 variants for at least 6 months in patients previously hospitalized for COVID-19. A weaker protection was, however, observed for the B.1.351 variant.

Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) has been shown to induce a humoral immune response with seroconversion occurring in most patients between 7 and 21 days after diagnosis [1, 2]. This early humoral response is mostly composed of immunoglobulin A (IgA), immunoglobulin M (IgM), and immunoglobulin G (IgG) directed against the viral surface glycoprotein spike (S), the nucleocapsid protein (NP), or the spike receptor-binding domain (RBD) [2]. The detection of such antibodies may reflect a neutralizing activity believed to be a key point in viral clearance [3, 4], as well as conferring a relative protection to the disease in the convalescent phase. Similar to other coronaviruses, anti–SARS-CoV-2 antibodies decline over time [5], which raised questions about the extent of the protection conferred and the potential risk of reinfection. Furthermore, the recent emergence of multiple SARS-CoV-2 variants raised additional questions on cross-reactivity of the acquired antibodies after coronavirus disease 2019 (COVID-19) [6].

Some publications have reported an association between the level of antibodies and the clinical severity, a higher level being observed in patients presenting the most critical form of the disease [4, 7–10]. However, if there is consistent evidence that outpatients usually develop weaker immune response, there are fewer data relating to patients in intensive care units (ICUs), a

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valuable population whose described antibody response could set the upper limit of the humoral immunity against SARS-CoV-2.

While vaccination is ongoing worldwide, the insufficient supply of doses makes prioritization strategies still needed. Being able to shape levels of immunity required to protect against severe reinfection would considerably assist public health strategies in this regard, in addition to being critical information to estimate if vaccines stand the test of time and emerging variants.

Our study explores the longitudinal evolution of antibody levels and of sera neutralizing activities in a French monocentric cohort of patients hospitalized for COVID-19 during the first wave of the SARS-CoV-2 pandemic and followed up for 6 months after hospital discharge. In addition to the ancestral viral strain, sera-neutralization activities against the emerging SARS-CoV-2 variants (B.1.1.7, B.1.351, and P.1) were evaluated.

MATERIALS AND METHODS

Cohort Description

We conducted a single-center prospective observational study on adult patients with laboratory-positive SARS-CoV-2 real-time reverse-transcription polymerase chain reaction admitted to Hôpital Européen Georges Pompidou (Paris, France) for at least 48 hours. All patients were initially enrolled from 17 March to 29 April 2020 and were then proposed for a clinical and serological follow-up at month 3 (M3) and month 6 (M6) postinfection. The study is part of the French Covid cohort (NCT04262921) sponsored by Inserm and was authorized by the French Ethics Committee CPP Ile-de-France VI (ID RCB:2020-A00256-33). This study was conducted with the understanding and the consent of each participant or their surrogate.

Data Collection

Demographic, clinical presentation, and comorbidity data during the index COVID-19 hospitalization were extracted from the electronic medical records collected in a standardized data collection form in the clinical data warehouse (CDW) of our hospital. The dedicated medical records were stored on an i2b2 platform in a CDW together with all other hospital health records.

Serological Assays

Abbott SARS-CoV-2 IgG assays (Des Plaines, Illinois) targeting SARS-CoV-2 nucleoprotein were done on an Architect i2000SR analyzer (Abbott), according to the manufacturer's instructions. Index value threshold for positivity was 1.4 as recommended. Beckman Coulter Access SARS-CoV-2 IgG assays (Brea, California) targeting the RBD of the SARS-CoV-2 spike surface protein were done on a UniCel DxI 800 Access Immunoassay System (Beckman Coulter), according to the manufacturer's instructions. Index value threshold for positivity was 1 as recommended. Qualitative results as well as index values were used for analysis for both assays.

Virus Strains

The ancestral non-D614G SARS-CoV-2 strain (BetaCoV/France/IDF0372/2020) was isolated in France from an imported case from Wuhan by the National Reference Center for Respiratory Viruses (NRC) hosted by Institut Pasteur. The reference D614G strain (hCoV-19/France/GE1973/2020) was supplied by the NRC. This viral strain was supplied through the European Virus Archive–Global platform. The B.1.1.7 strain originated from an individual in Tours (France) who returned from the United Kingdom [11]. The B.1.351 strain (CNR 202100078) originated from an individual in Créteil (France) [11]. The P.1. strain (TY7-501), first identified in Brazil, was obtained from Global Health security action group Laboratory Network (GISAID sample ID: hCoV-19/Japan/TY7-501/2021; GISAID ID: EPI_ISL_833366). Individuals provided informed consent for the use of their biological materials. The variant strains were isolated from nasal swabs on Vero E6 cells and amplified by 1 or 2 passages. Of note, the sequence of TY7-501 contains a G181V mutation on its S protein compared with the original clinical sample sequence. Titration of viral stocks was performed on Vero E6 cells, with a limiting dilution technique allowing a calculation of the 50% tissue culture infectious dose, or on S-Fuse cells [11]. Viruses were sequenced directly on nasal swabs and after 1 or 2 passages on Vero cells.

S-Fuse Neutralization Assay

Neutralization was performed using the S-Fuse reporter system, as previously described [11]. In brief, U2OS-ACE2 GFP1–10 or GFP11 cells, which become GFP^+ upon infection with SARS-CoV-2, were mixed (1:1 ratio) and plated at 8×10^3 cells per well in a µClear 96-well plate (Greiner Bio-One). SARS-CoV-2 strains were incubated with sera at the indicated dilutions for 15 minutes at room temperature and added to S-Fuse cells. All sera were heat inactivated for 30 minutes at 56°C before use. After 18 hours of incubation at 37°C 5% carbon dioxide, cells were fixed with 2% paraformaldehyde, washed, and stained with Hoechst (1:1000 dilution; Invitrogen). Images were acquired on an Opera Phenix high-content confocal microscope (PerkinElmer). The GFP area, the number of syncytia, and nuclei were quantified using Harmony software (PerkinElmer). The percentage of neutralization was calculated using the number of syncytia with the following formula: 100 × 1 – (value with serum – value in "noninfected") / (value in "no serum" – value in "noninfected"). Neutralizing activity of each sera was expressed as the ID_{50}, calculated using the percentage of neutralization at each different concentration. Cells were tested negative for *Mycoplasma*. Neutralization determined with the S-Fuse reporter system correlates to pseudovirus neutralization assay [12].

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Statistical Analyses

Statistics were performed using NCSS 2012 software (G Hintze, Kaysville, Utah). All numerical data were checked for normality and nonnormal distributions were transformed (using ExpNorScore function on NCSS, which returns the expected value of the normal order statistic corresponding to X).

Continuous variables are reported as mean (standard deviations). Discrete variables are described as counts and percentages. Groups were compared using 2-sample *t* test or Wilcoxon rank test when necessary for continuous variables and Fisher exact test or χ^2 for discrete variables.

We also performed a multiple regression analysis to assess variables correlated to the seroneutralization at M6. For analyses, *P* values < .05 were considered significant.

RESULTS

Between 17 March and 29 April 2020, 354 patients were hospitalized at Hôpital Européen Georges Pompidou with a confirmed SARS-CoV-2–positive pneumonia. By 16 November 2020, 85 deaths (24.0%) had occurred during or after hospitalization. From 17 June to 16 November 2020, we were able to manage a complete follow-up for 107 of these patients with 2 time-point visits at 3 and 6 months after hospital discharge (Supplementary Figure 1). The M3 and M6 visits were performed with a median interval of 98 days (interquartile range [IQR], 91–101 days) and 203 days (IQR, 191–216 days), respectively. Among those patients with a complete follow-up, 32.7% (35/107) had required medical care in the ICU during the acute COVID-19 phase; 15 of them (15/35 [42.9%]) required invasive mechanical ventilation (MV). The oxygenation maximal flow in patients in non-ICU, in patients not requiring MV in the ICU, and in patients before MV in ICU was 2.0 ± 1.6, 10.7 ± 5.2, and 13.0 ± 3.6 L/minute, respectively (*P* < .0001). A majority of patients were male (73/107 [68.2%]) with a mean age of 58.7 ± 14.0 years. A history of cardiovascular risk factors (chronic cardiac disease, diabetes, obesity, hypertension, chronic kidney disease) was found in 51.4% (55/107), and 10.2% (11/107) had immunosuppressive diseases (cirrhosis, asplenia, sickle cell anemia, solid organ or stem cell transplantation, human immunodeficiency virus infection, primary immunodeficiency, chronic hematological disease, malignant neoplasm, autoimmune disorder). Moreover, 5.6% (6/107) of them were previously treated with immunosuppressive therapy. All patients' characteristics are listed in Table 1.

Table 1. Demographic, Comorbidities, and Treatment During the Acute Phase

Characteristic	Overall (N = 107)	ICU (n = 35)	No ICU (n = 72)	*P* Value
Age, y, mean ± SD	58.7 ± 14.0	59.0 ± 11.9	58.5 ± 15.0	.86
Female sex	34 (31.8)	10 (28.6)	24 (33.3)	.61
BMI, kg/m^2, mean ± SD	27.2 ± 4.5	27.6 ± 4.0	27.0 ± 4.7	.46
Medical history				
Previous heart disease	14 (13.1)	4 (11.4)	10 (13.9)	1.00
Hypertension	36 (33.6)	14 (40.0)	22 (30.6)	.33
Diabetes	17 (15.9)	8 (22.9)	9 (12.5)	.17
Chronic renal disease	8 (7.5)	2 (5.7)	6 (8.3)	1.00
Chronic pulmonary disease	10 (9.3)	2 (5.7)	8 (11.1)	.49
Asthma	12 (11.2)	2 (5.7)	10 (13.9)	.33
Hepatic disease	2 (1.9)	1 (2.9)	1 (1.4)	.55
Active cancer	4 (3.7)	1 (2.9)	3 (4.2)	1.00
Hematologic disease	7 (6.5)	2 (5.7)	5 (6.9)	1.00
HIV	2 (1.9)	0 (0.0)	2 (2.8)	1.00
Rheumatic disease	5 (4.7)	2 (5.7)	3 (4.2)	.72
Sickle cell disease	2 (1.9)	0 (0.0)	2 (2.8)	.32
Immunosuppressive treatment	6 (5.6)	3 (8.6)	3 (4.2)	.39
COVID-19 treatment				
Oxygenation maximal flow, L/min, mean ± SD	4.9 ± 5.0	10.8 ± 4.3	2.0 ± 1.6	**<.0001**
Invasive mechanical ventilation	15 (14.2)	15 (42.9)	0 (0.0)	**<.0001**
Antiviral therapy[a]	2 (1.9)	2 (5.7)	0 (0.0)	.10
Anti–IL-6 antibody	11 (10.3)	10 (28.6)	1 (1.4)	**<.0001**
Anti–IL-1 antibody	1 (0.9)	1 (2.9)	0 (0.0)	.32
Hydroxychloroquine	11 (10.3)	6 (17.1)	5 (6.9)	.10
Corticosteroïds	3 (2.8)	3 (8.6)	0 (0.0)	**.03**

Data are presented as No. (%) unless otherwise indicated. *P* values less than 0.05 in bold are statistically significant.

Abbreviations: BMI, body mass index; HIV, human immunodeficiency virus; ICU, intensive care unit; IL-1, interleukin 1; IL-6, interleukin 6; SD, standard deviation.

[a]Antiviral therapy: lopinavir for 1 patient and remdesivir for 1 patient.

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Longitudinal Serological Assays at 3 and 6 Months Postinfection

We quantitively assessed the presence of IgG recognizing the NP or S domain in serum samples from these 107 patients. At M3 after COVID-19 infection, anti-S and anti-NP antibodies were detected in all patients. Both anti-S and anti-NP IgG levels significantly decreased between M3 and M6 (respectively, 17.1 ± 13.6 to 7.8 ± 9.2, $P < .0001$ and 6.6 ± 2.3 to 3.8 ± 2.4, $P < .0001$; Figure 1). Six months after COVID-19 infection, anti-S serology was considered negative (ie, antibodies below the commercial threshold) in 9 samples (9/107 [8.4%]) and anti-NP in 17 samples (17/107 [15.9%]). However, only 3 patients had a negative serological status with both assays (3/107 [2.8%]) at M6.

Neutralizing Activities Against Ancestral SARS-CoV-2 at 3 and 6 Months Postinfection

We then aimed to assess if a persistent serum neutralizing activity was detected up to 6 months following COVID-19 infection, independent of anti-S or anti-NP levels. To do so, we used S-Fuse cells (specifically designed to become GFP^+ when productively infected by SARS-CoV-2) to evaluate our patients' sera propensity to prevent such infection. At minimum dilution (1/30), seroneutralization was observed in all samples at M3 and M6, even when anti-S and anti-NP IgG were considered as negative regarding the commercial kit threshold.

We next quantified seroneutralization by performing serial dilutions in order to define the ID_{50} neutralization (maximum dilution to maintain a 50% neutralization capacity). ID_{50} neutralization significantly decreased between M3 and M6 (Figure 2), with residual values nevertheless indicating a high neutralizing activity at M6.

Impact of the Initial Clinical Severity on Residual Humoral Immunity

As all measures were highly variable between patients, we then sought factors associated with higher levels of neutralizing activities. In a multiple regression analysis, we found that initial management in ICU and the need for an invasive MV were the only 2 factors significantly associated with a higher rate of ID_{50} neutralization at M6 (Table 2). When considering ID_{50} neutralization according to ICU hospitalization and the need for MV, we found that patients in ICU had significantly higher neutralizing activities as compared to non-ICU patients, with the highest levels observed in ICU patients who had invasive MV (Figure 3A). A similar trend was observed at M3, however not reaching significance (Figure 3A).

We then observed that anti-S IgG modestly correlated with sera ID_{50} neutralization ($R = 0.54$ [95% confidence interval {CI}, .39–.67]; $P < .0001$), with a weaker association for anti-NP IgG ($R = 0.33$ [95% CI, .25–.49]; P .0007).

We then analyzed anti-S and anti-NP IgG levels at M3 and M6 according to the initial management in ICU and the need for invasive MV. We found higher levels of anti-NP IgG in mechanically ventilated patients in ICU vs non–mechanically ventilated patients admitted to ICU vs patients in non-ICU medical departments at M3 (7.6 ± 1.8 vs 7.2 ± 2.2 vs 6.2 ± 2.3, $P = .03$ by Kruskal-Wallis test; Figure 3B), with a significantly higher rate in ICU patients vs non-ICU patients (7.4 ± 2.0 vs 6.2 ± 2.3, $P = .01$). We confirmed this result at M6 (5.6 ± 1.8. vs 4.4 ± 2.0 vs 3.3 ± 2.4, $P = .0005$ by Kruskal-Wallis test; Figure 3B), with a significantly higher rate in ICU patients vs no ICU patients

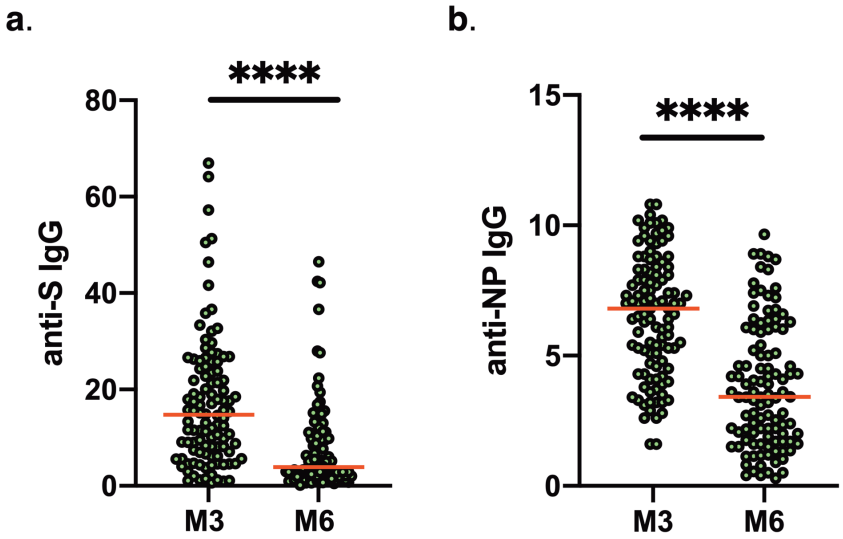

Figure 1. Temporal evolution of anti–severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) antibodies. *A* and *B*, Level of anti-spike (S) and anti-nucleocapsid (NP) immunoglobulin G (IgG) in the cohort, 3 months (M3) and 6 months (M6) after hospitalization for reverse-transcription polymerase chain reaction–confirmed SARS-CoV-2 infection. The IgG level was determined quantitatively by chemiluminescent microparticle immunoassay. Differences between time-points were analyzed with *t* test, ****$P < .0001$. The red line represents the median for each time-point.

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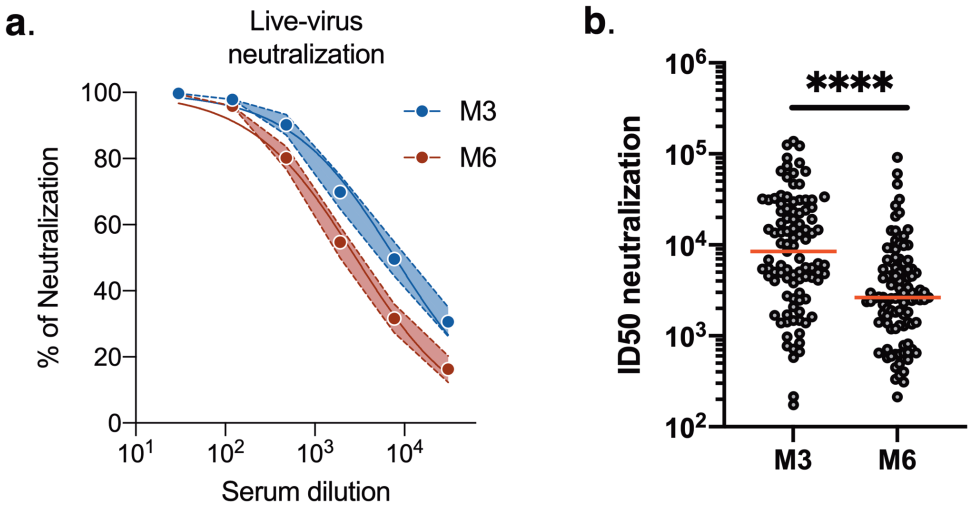

Figure 2. Temporal evolution of neutralizing activity *A*, Sera collected 3 (blue) and 6 (red) months (M3 and M6, respectively) posthospitalization were tested with serial dilutions for their ability to neutralize severe acute respiratory syndrome coronavirus 2. Shown is the mean activity at each serum dilution with the 95% confidence interval. *B*, The inhibitory dose of 50% (ID_{50}) neutralization is depicted at M3 and M6 after hospitalization. Statistical analysis: Wilcoxon tests were performed. ****$P < .0001$. The red line represents the median for each time-point.

(4.9 ± 2.0 vs 3.3 ± 2.4, $P = .001$). In contrast, this pattern was not observed with the anti-S antibodies as there were no significant differences between the ICU vs the non-ICU groups at M3 and at M6 (Figure 3C).

Neutralizing Activities Against SARS-CoV-2 Emerging Variants

We then used sera collected at 6 months following COVID-19 infection to assess their neutralizing activities against multiple variants, including D614G, B.1.1.7, B.1.351, and P.1 variants.

Table 2. Multiple Regression Analysis of Variables Associated With Seroneutralization at 6 Months

Variable	Adjusted[a] Regression Coefficient ± SE	*P* Value
ICU hospitalization	0.372 ± 0.099	**.0003**
Body mass index	0.036 ± 0.021	.10
Preexisting heart disease	0.1064 ± 0.1466	.47
Hypertension	0.1397 ± 0.1087	.20
Diabetes	0.129 ± 0.1339	.34
Chronic pulmonary disease	–0.0259 ± 0.1718	.88
Immune disorders	0.002 94 ± 0.1374	.98
Invasive mechanical ventilation	0.4155 ± 0.1352	**.0028**
Anti–IL-6 antibody	0.2215 ± 0.1591	.17
Corticosteroïds	0.2538 ± 0.2938	.39
Serology		
3 mo anti-NP	0.3368 ± 0.0973	**.0008**
3 mo anti-spike	0.4662 ± 0.0923	**<.00001**
6 mo anti-NP	0.358 57 ± 0.0960	**.0003**
6 mo anti-spike	0.6434 ± 0.0899	**<.0001**

P values less than 0.05 in bold are statistically significant.

Abbreviations: ICU, intensive care unit; IL-6, interleukin 6; NP, nucleocapsid protein; SE, standard error.

[a]Adjusted for age and sex.

At minimum dilution (1/30), seroneutralization was observed in all samples against the D614G and B.1.1.7 variants. We then quantified seroneutralization by performing serial dilutions to define the ID_{50} neutralization for the 4 strains. The D614G, B.1.1.7, and P.1 strains were similarly sensitive to the sera with a high neutralizing activity observed for these 3 variants. In contrast, the neutralization titers against B.1.351 were significantly lower with a 3-fold decrease in ID_{50} between the D614G and B.1.351 strains ($P < .0001$; Figure 4A). ID_{50} for the B.1.351 variant was also significantly lower as compared to that for the B.1.1.7 and P.1 variants (Figure 4B).

DISCUSSION

In the present study, we described the longitudinal evolution of IgG levels and seroneutralization at 3 and 6 months postinfection in a relatively large prospective cohort of 107 hospitalized patients—with a third of severe cases tending to be scarce in literature—and thus provides important data about the evolution of humoral immunity after a hospitalization for COVID-19 infection. We found that at least 1 serology assay was still positive at 6 months in 97.2% of the studied patients. Although antibody levels decreased significantly over time, with rates dropping under the positivity threshold in a few cases, all patients' sera conserved an effective neutralizing activity at 6 months postinfection against the ancestral strain. Seroneutralization remained higher at 3 and 6 months in patients who had required intensive care. We also used our sera collection to estimate the levels of humoral protection against the emerging SARS-CoV-2 variants. In these additional in vitro experiments, we found that sera-neutralizing activities was also effective against the B.1.1.7 and P.1 variants (also known as the

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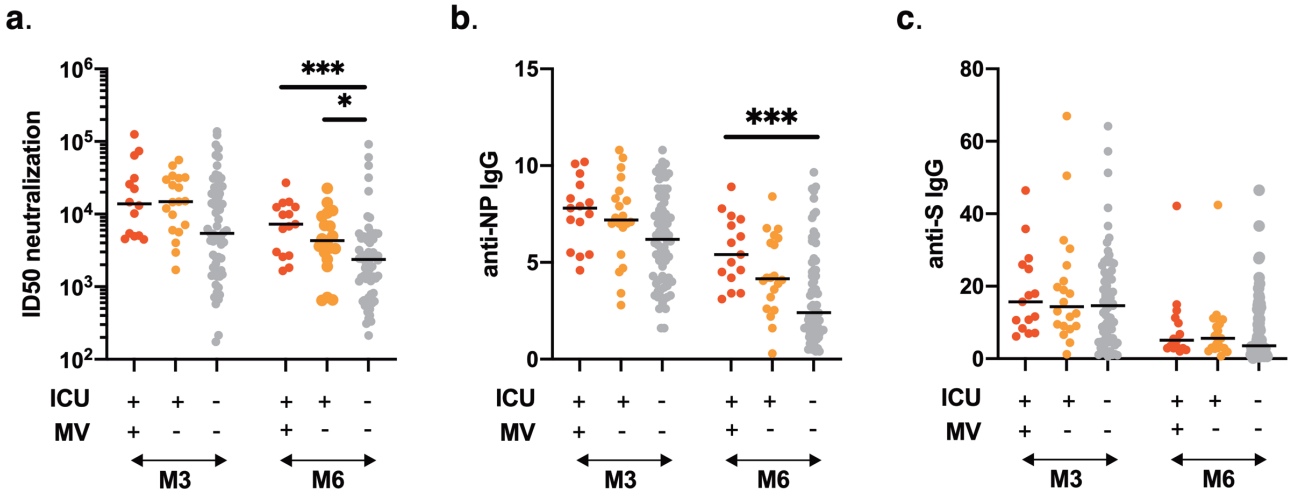

Figure 3. Evaluation of the humoral response according to the initial clinical presentation. Inhibitory dose of 50% (ID_{50}) neutralization (*A*), anti-nucleocapsid (NP) immunoglobulin G (IgG) (*B*), and anti-spike (S) IgG (*C*), at 3 and 6 months postinfection (M3 and M6, respectively), in 3 different subgroups of the cohort: patients requiring mechanical ventilation (MV) (+/+), patients requiring intensive care unit (ICU) admission without MV (+/–), and patients hospitalized in conventional care (–/–). The IgG level was determined quantitatively by chemiluminescent microparticle immunoassay. ID_{50} was determined by serial dilution. Differences between the 3 subgroups in each time-point were analyzed with 1-way analysis of variance followed by multiple comparisons. $^{*}P < .05$, $^{***}P < .001$ for comparison between groups. The black line represents the median for each time-point.

UK and Brazilian variants), but was potentially weaker for the B.1.351 strain (also known as South African variant).

Higher ID_{50} against the ancestral strain were observed in patients with more severe presentations, even at distance of infection. This correlation between seroneutralization and clinical severity has been previously described [7, 10, 13, 14], and our results now indicate that this trend might persist over time. Interestingly, we found that anti-NP IgG titers were higher according to the stage of severity, which was not observed for anti-S IgG. Early after symptom onset, anti-NP response had already been reported as a possible marker of severity, associated to delayed viral clearance and disease severity [8]. Whether this exacerbated humoral response in severe patients is a protective adaptation to a more intense viral load or if it plays a putative role in pathogenicity remains subject to debate [15, 16].

At M6 postinfection, we found that anti-S IgG titers correlated with sera ID_{50} neutralization, but not anti-NP IgG. This was generally in line with other works that had underlined,

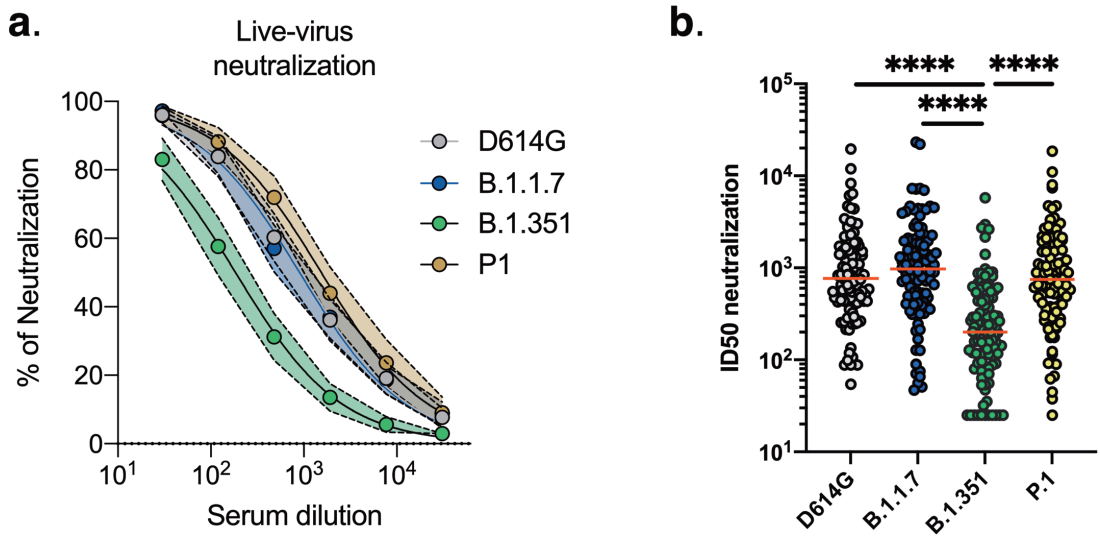

Figure 4. Neutralizing activity against D614G, B.1.1.7, B.1.351, and P.1 variants. *A*, Sera collected 6 months posthospitalization were serially diluted and tested for their ability to neutralize the 4 indicated severe acute respiratory syndrome coronavirus 2 strains (D614G, B.1.1.7, B.1.351, and P.1). Shown is the mean activity at each serum dilution with the 95% confidence interval. *B*, Inhibitory dose of 50% (ID_{50}) neutralization for the 4 viral isolates. Each dot represents an individual. Statistical analysis: 1-way analysis of variance, followed by multiple comparisons between groups. $^{****}P < .0001$. The red line represents the median for each time-point.

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at different times postinfection, a relatively strong correlation between neutralizing antibodies and anti-S or anti-RBD antibodies, and a usually poorer correlation to anti-NP antibodies [9, 14, 17, 18].

The evolution of sera-neutralization over time and in response to emerging variants is one important element to consider when questioning the extent of effective protection conferred by a priori infection and thus helps in evaluating the strength of shield immunity during this pandemic. In line with first encouraging results [11, 13, 17, 19–21], we confirmed in this study the persistence of neutralization up to 6 months postinfection against the ancestral strain, but also the existence of broader and similarly effective neutralizing activities against novel variants including B.1.1.7 and P.1. These results are in favor of antibody cross-reactivity and potential protection against reinfection with these variants. As compared to other strains, we observed a weaker protection against the B.1.351 variant, however, with a substantial neutralizing activity observed in most of the patients. These data suggest that these antibodies acquired during a prior COVID-19 infection might not confer a complete protection against this emerging variant first described in South African patients [22]. We took benefit of the available sera collection and the development of a novel assay to estimate these activities but cannot extrapolate on a higher risk of reinfection in these convalescent patients. Several publications based on pseudovirus or virus neutralizing assays outlined that variants could partially evade humoral immunity—in exposed patients as well as vaccinated individuals [23–25].

Our cohort of patients with the most critical forms of COVID-19 represents a valuable population to explore maximal antibody response and set the upper limit of humoral immunity against SARS-CoV-2. Our results suggest that our patients should be protected at least 6 months against future reinfection. So far, there are few cases of reinfection published in the literature [26]. Reinfection rates have been estimated as low in large recent observational studies despite waning neutralizing antibodies [27, 28]. However, it is impossible to extrapolate on future infections with novel variants. Interestingly, a recent model predicted a relationship between neutralizing levels and immune protection against the ancestral strain and novel variants, as well as a protection against severe disease [29]. Further studies are nonetheless required, especially regarding the B.1.351 variant, to determine if partial humoral escape can clinically lead to severe events. Cellular immunity also appears as a major shield against SARS-CoV-2, with the development of durable T memory cells [20], whose reactivity could be only slightly impacted by variants [30].

Overall, patients who survived the most critical forms of COVID-19 consequently developed an intense and prolonged humoral immunity. These levels tend to correlate with the severity of the initial presentation, with patients in ICU who had invasive MV having the highest neutralizing activities as compared to ICU and non-ICU patients. The developed antibodies are cross-reacting and should confer a similar protection against emerging variants, with the notable exception of the B.1.351 variant. This pattern may be considered in global deployment of vaccination worldwide. Future studies will establish whether acquired antibodies also protect against reinfections with the emerging SARS-CoV-2 variants. Our data indicate the need for a specific attention to the B.1.351 variant.

Notes

Author contributions. J. S. H., M. B., M. L., D. V., and E. T. designed the study. J. S. H., M. L., and A. F. had full access to all of the data, and take the responsibility of data integrity and accuracy of the data analysis. J. S. H. drafted the manuscript with the help of M. B., M. L., A. F., and D. P., and B. M. A. F. and J. S. H. performed all the analyses. M. B., M. L., B. V., N. R., A. B., H. P., D. L., and D. V. collected the data. D. P., B. M., T. B., and O. S. performed the neutralization experiments. All authors critically revised the manuscript for important intellectual content and gave final approval for the version to be published.

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APPENDIX

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